

Projectplan: Developing an AI-based predictive model for drought and flood early warning systems in African agriculture for Lidi Smart Solutions

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1. Business and problem context

Lidi Smart Solutions is an organization, based in The Hague, the Netherlands, involved in several projects funded by the European Union. The company's core activities focus on digital innovation, artificial intelligence (AI), and sustainability.

This specific assignment is being carried out for the *AI4AgriBizAfrica* concept, which is currently under development within the Horizon Europe framework. The objective of this concept is to improve climate and weather models in Africa by leveraging AI and open data technologies (Horizon Europe, 2026). Within this domain, the focus is on the vulnerability of African agriculture to climate variability. Smallholder farmers are heavily dependent on predictable rainfall. Consequently, extreme weather conditions can lead to crop failure and uncertainty access to food (Muzata, n.d.; Omotoso & Ayodele, 2026).

This project utilizes open climate reanalysis data from the Copernicus Climate Data Store (ERA5). This dataset consists of six atmospheric parameters measured on an hourly basis across the African continent (*ERA5: Data Documentation - Copernicus Knowledge Base - ECMWF Confluence Wiki*, 2026). This data will be made available via a Microsoft Azure cloud storage environment.

2. Background and problem definition

The motivation for this assignment is the urgent need to strengthen agricultural resilience in Africa and support farmers' decision-making. Climate disasters, such as droughts and floods, cause significant economic damage and threaten livelihoods (Reed et al., 2022; Balgah et al., 2023). Timely and reliable information provided through Early Warning Systems (EWS) can help farmers make better decisions regarding planting, irrigation, and risk mitigation (CWP, 2025). To support these systems, there is a clear need for a predictive AI model.

The core problem is that early signals of drought and flooding are often hidden within complex atmospheric patterns. Large volumes of available climate data remain untapped, and local institutions lack access to AI-driven tools for early warnings. Currently, there is a lack of a functional model capable of translating meteorological parameters into actionable risk indicators.

3. Goals

After finishing the assignment there will be a technically robust proof-of-concept model which can predict the chance of onset drought and floods based on ERA5 climate parameters. With this working prototype non-technical stakeholders (like agriculture advisors and policy makers) will find easy to understand visualisations about the weather data. This way the organization is in state to give the *AI4AgriBizAfrica* initiative shape and help prepare local communities for climate risks.

4. Provisional research question and sub-questions

How can we use hourly ERA5 atmospheric data to build a machine learning model that predicts the probability of drought and flooding events in specific African regions?

Phase (Crisp-DM)	Sub question	Methods
<i>Business understanding</i>	<i>How can predictive machine learning model contribute to a reliable and practical warning system for droughts and floods in African agricultural regions?</i>	<i>Literary research (Baarda & Van Vianen, 2021, pp. 64–78)</i> <i>Stakeholder analysis</i> <i>Problem definition</i> <i>Desk research</i> <i>Definition of success criteria</i>
<i>Data understanding</i>	<i>What are the characteristics, quality and limitations of ERA5 climate dataset?</i> <i>To what extend is the dataset suitable for predicting drought and flood risk?</i>	<i>Exploratory Data Analysis (Jha, 2025)</i> <i>Data validation (Prometeia, 2025)</i>
<i>Data preparation</i>	<i>How can the raw ERA5 climate variables be transformed into meaningful, model-ready features that can represent drought and flood risk?</i>	<i>Feature engineering (Francis, 2024)</i> <i>Feature selection (Belcic & Stryker, n.d.)</i> <i>Normalization / standardization (AbhijeetDataVision, 2025)</i> <i>Handling class imbalance (Nandi, 2025)</i> <i>Time-based train/test split (Chalokia, 2021)</i>
<i>Modelling</i>	<i>Which machine learning models provide the best balance between reliable and interpretable predictions of drought and flood risk?</i>	<i>Model comparison (Devansh, 2023)</i> <i>Hyperparameter tuning (Singh, 2024)</i> <i>Performance evaluation (Caballar & Stryker, n.d.)</i> <i>Overfitting / underfitting analysis (Stuart, 2025)</i> <i>Feature importance analysis (Bhuva, 2025)</i>

<i>Evaluation</i>	<i>To what extent does the developed model meet the technical, societal and practical objectives of the project?</i>	<i>Assessment of model performance against predefined project goals and success criteria</i> <i>Analysis of generalizability by comparing performance across different time periods</i> <i>Analysis of error types (Relangi, 2024)</i> <i>Scenario analysis</i> <i>Ethical evaluation</i> <i>Reflection on dataset limitations and model assumptions</i> <i>Writing a final report (Baarda & Van Vianen, 2021, pp. 279–302)</i>
<i>Deployment</i>	<i>N.v.t.</i>	<i>N.v.t.</i>

5. Planning

Phase	Method	Intermediate product	Time
<i>Analysing the climate variables related to onset drought and floods.</i>	<i>Literary-analysis</i>	<i>Literary report and justify feature selection.</i>	<i>Week 1-7</i>
<i>Obtain and prepare ERA5 climate data.</i>	<i>Data loading via MS Azure, data cleaning in Python.</i>	<i>Clean dataset.</i>	<i>Week 5-9</i>
<i>Transforming meteorological parameters to significant features.</i>	<i>Feature engineering techniques.</i>	<i>Dataset including derived and defined target variables.</i>	<i>Week 8-12</i>
<i>Justify model choices and training.</i>	<i>Apply model training using.</i>	<i>Trained ML-models</i>	<i>Week 10-14</i>
<i>Evaluate model results, robustness, and reliability.</i>	<i>Validation techniques and evaluation metrics.</i>	<i>Evaluation report with model comparison and feature importance.</i>	<i>Week 13-17</i>
<i>Create visualisation and ethical reflection and final report.</i>	<i>Data visualization and structured report.</i>	<i>Final report with proof-of-concept and presentation.</i>	<i>Week 14-18</i>

6. Risk Analysis

Within this project, a machine learning model is developed to predict drought and flood risks in African regions based on ERA5 climate data. The project focuses on the use of large-scale, time-dependent datasets and the necessity to transform complex atmospheric variables into meaningful indicators for climate-related events.

Such data science projects naturally involve various uncertainties, both on a technical, methodological, and organizational level. Limitations in data quality and availability, challenges in defining appropriate target variables, and complexities in model validation can directly affect the reliability and generalizability of the model. In addition, the limited project duration plays an important role in balancing depth, accuracy, and practicality.

Systematically identifying and managing these risks is therefore essential to ensure the quality and usability of the project outcomes and to enable timely adjustments throughout the project. This risk analysis aims to identify the most relevant risks, assess their likelihood and impact, and define appropriate preventive strategies. This allows for proactive management and increases the likelihood of achieving a successful and relevant final result.

Risk	Likelihood	Impact	Description	Preventive measures
Insufficient data quality.	Medium	High	Climate data may contain missing values, noise, and inconsistencies, leading to unreliable features and reduced model performance.	Perform exploratory data analysis (EDA), apply data cleaning and imputation, and validate and normalize the data.
Complexity of feature engineering.	High	High	Extracting relevant climate indicators (e.g., drought periods or precipitation sums) requires domain knowledge. Poor features result in weak model performance.	Apply iterative feature development and support it with extensive literature research.
Unclear target definition.	Medium	High	Droughts and floods are not directly observable, leading to subjective definitions.	Define multiple target combinations based on literature and compare their effectiveness.
Insufficient model performance.	Medium	High	Models may not achieve satisfactory and desired performance.	Compare multiple algorithms, optimize hyperparameters, and use appropriate evaluation metrics.
Overfitting due to time dependency.	High	Medium	Time series data can lead to data leakage when using incorrect validation strategies.	Apply time-based splits and cross-validation methods that respect temporal order.

Data access issue (cloud).	Low	Medium	Potential limitations in accessing the Azure environment.	Use local copies or caching mechanisms.
Time constraints within project duration.	High	High	The project involves multiple complex components within a limited timeframe, potentially reducing depth, quality or succes of the project.	Work iteratively with clear prioritization.
Dependency on the stakeholders.	Low	Medium	Limited availability of feedback may result in misalignment with expectations.	Schedule regular standup meetings and share intermediate results.
Limited model interpretability.	Medium	Medium	Complex models are difficult to interpret, which may reduce stakeholder trust.	Use explainability techniques such as feature importance and SHAP.
Limited practical applicability.	Medium	High	Results may not align well with real-world applications of the AI solution.	Focus on understandable outputs (visualizations and risk scores) and relevant use cases.
Ethics and bias in data.	Low	Medium	Potential bias in data or model results may lead to incorrect decision-making when using the AI application.	Be transparent about limitations, reflect on bias and explain the tool as a decision-support system.
Incorret or missing references.	Medium	Medium	Insufficient or incorrect referencing of literature and datasets may reduce reliability and violate academic guidelines.	Apply APA guidelines, systematically track sources and verify references

6.1 Analysis of critical risks

The major risks within this project are related to data quality, feature engineering, model validation, and project management.

Firstly, the quality of the climate data represents a major risk. Incomplete or inconsistent data can directly introduce noise into the features, which negatively affects the performance and reliability of the model.

Secondly, feature engineering is a critical step. Transforming raw climate data into relevant features requires both technical expertise and domain knowledge of climate processes. Insufficiently supported features may result in the model failing to capture important patterns.

Model validation in the context of time series data also requires careful attention. Incorrect validation methods can lead to data leakage, causing model performance to appear overly optimistic and resulting in poor real-world performance. Proper handling of the train-test split is therefore essential for a realistic evaluation.

Furthermore, time constraints play a significant role within this project. Given the limited timeframe, the complexity of data processing, modelling, and interpretation may result in reduced depth or quality.

Finally, the practical applicability and interpretability of the model are important considerations. A model may perform well from a technical perspective, but still have limited value if the results are not understandable or usable for stakeholders.

Conclusion

The identified risks are largely inherent to machine learning projects involving complex and large-scale datasets. These risks mainly relate to data quality, model development, and the translation of technical results into practical applications.

By applying appropriate methods and techniques, such as iterative development, correct validation strategies, and well-founded feature engineering, these risks can be effectively managed. In addition, attention to interpretability and transparency contributes to the usability and reliability of the model.

Although not all risks can be eliminated, a structured and goal-oriented approach can significantly reduce their impact. This increases the likelihood that the project will result in a successful proof-of-concept for predicting drought and flood risks.

7. Final qualifications

<i>Final qualifications</i>	<i>Activities</i>
<i>Problem analysis</i>	<i>Execution of literary research about climate variables to define the problem domain.</i>
<i>Data gathering and -preparation</i>	<i>Apply feature engineering to time-series data.</i>

<i>Modelling</i>	<i>Training, testing and comparing of ML models bases on validation techniques and metrics</i>
<i>Reporting and advising</i>	<i>Presentation of finding using clear visualisations for non-technical stakeholders and advising about model use.</i>
<i>Investigative and reflective attitude and skills</i>	<i>Reflecting on potential bias, regional honesty and responsible use of AI within climate issues.</i>

8. Bibliografie (volgens APA)

Horizon Europe. (2026, February 17). Research and Innovation. [https://research-and-](https://research-and-innovation.ec.europa.eu/funding/funding-opportunities/funding-programmes-and-open-calls/horizon-europe_en)

[innovation.ec.europa.eu/funding/funding-opportunities/funding-programmes-and-open-calls/horizon-europe_en](https://research-and-innovation.ec.europa.eu/funding/funding-opportunities/funding-programmes-and-open-calls/horizon-europe_en)

Muzata, S. (n.d.). *Africa's smallholder farmers face collapse if we do not act on climate change.*

CGIAR System. <https://www.cgiar.org/news-events/news/africas-smallholder-farmers-face-collapse-if-we-do-not-act-on-climate-change>

Omotoso, A., & Ayodele, M. (2026). Nigerian farmers talk about how climate change is affecting staple food crops – and what can help. *The Conversation*.

<https://doi.org/10.64628/aaj.9xg5jra6s>

ERA5: data documentation - Copernicus Knowledge Base - ECMWF Confluence Wiki. (2026,

January 5). <https://confluence.ecmwf.int/display/CKB/ERA5%3A+data+documentation>

Reed, C., Anderson, W., Kruczkiewicz, A., Nakamura, J., Gallo, D., Seager, R., & McDermid, S. S. (2022). The impact of flooding on food security across Africa. *Proceedings of the National Academy of Sciences*, 119(43), e2119399119.

<https://doi.org/10.1073/pnas.2119399119>

Balgah, R. A., Ngwa, K. A., Buchenrieder, G. R., & Kimengsi, J. N. (2023). Impacts of Floods on Agriculture-Dependent Livelihoods in Sub-Saharan Africa: An Assessment from Multiple Geo-Ecological Zones. *Land*, 12(2), 334. <https://doi.org/10.3390/land12020334>

CWP. (2025, November 10). *Early Warning Systems in Africa* - CIMA Research Foundation.

CIMA Research Foundation. <https://www.cimafoundation.org/en/focus/early-warning-systems/africa/>

Singh, A. (2024, December 10). *Hyperparameter tuning: Beyond the basics*. Medium.

<https://medium.com/@abhaysingh71711/hyperparameter-tuning-beyond-the-basics-34d36b01448>

- Devansh. (2023, April 25). *How to properly compare machine learning models*. Artificial Intelligence Made Simple. <https://www.artificialintelligencemadesimple.com/p/how-to-properly-compare-machine-learning>
- Caballar, R. D., & Stryker, C. (n.d.). *What is model performance in machine learning?* IBM Think. <https://www.ibm.com/think/topics/model-performance>
- Chalokia, M. (2021, May 2). *Time-based splitting and determining if train & test data come from the same distribution*. Medium. <https://medium.com/@manrajchalokia/time-based-splitting-and-determining-if-train-test-data-come-from-the-same-distribution-e1d2ea881af8>
- Nandi, S. (2025, January 13). *How to handle class imbalance in machine learning*. Medium. <https://medium.com/@sushmita2310/how-to-handle-class-imbalance-in-machine-learning-c4eaf8a26026>
- AbhijeetDataVision. (2025). *Data normalization vs standardization: When and why it matters in analysis*. Medium. <https://medium.com/@AbhijeetDataVision/data-normalization-vs-standardization-when-and-why-it-matters-in-analysis-e8f7702fe0af>
- Stuart, K. (2025, December 11). *The balancing act: Understanding overfitting and underfitting in machine learning*. Medium. <https://medium.com/@kaylenstuart/the-balancing-act-understanding-overfitting-and-underfitting-in-machine-learning-edae067d12f9>
- Bhuva, L. (2025, March 20). *Understanding feature importance in machine learning*. Medium. <https://medium.com/@lomashbhuva/understanding-feature-importance-in-machine-learning-d86ec50e0055>
- Belcic, I., & Stryker, C. (n.d.). *What is feature selection?* IBM Think. <https://www.ibm.com/think/topics/feature-selection>
- Francis, S. F. (2024, October 5). *The importance of feature engineering in data science*. Medium. <https://medium.com/@silva.f.francis/the-importance-of-feature-engineering-in-data-science-11c45f0f03b0>

Jha, G. (2025, September 1). *A holistic guide to exploratory data analysis (EDA) for machine learning and deep learning*. Medium. <https://medium.com/@post.gourang/a-holistic-guide-to-exploratory-data-analysis-eda-for-machine-learning-and-deep-learning-bc4f18f0143b>

Prometeia. (2025, February 3). *The importance of data quality and validation in machine learning*. Prometeia. <https://prometeia.com/en/about-us/insights/article/the-importance-of-data-quality-and-validation-in-machine-learning>

Relangi, K. S. (2024, June 29). *Understanding model evaluation metrics for machine learning*. Medium. <https://medium.com/@kavyasrirelangi100/understanding-model-evaluation-metrics-for-machine-learning-160d385b72c5>

Baarda, B., & Van Vianen, R. (2021). *Dit is onderzoek!* (4e ed.). Noordhoff.